

**SIDDHARTHA ACADEMY OF HIGHER EDUCATION
(Deemed to be University)**

**VELAGAPUDI RAMAKRISHNA SIDDHARTHA SCHOOL OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**

B.Tech III Year, ACADEMIC YEAR: 2026-27

24CS381-ENGINEERING PROJECTS FOR COMMUNITY SERVICES(EPICS)

B.Tech (Computer Science & Engineering)-CSE

Review 0: Template

Course:CSE

Section:CSE-C

Batch. No:23

Name of the Students and Roll. No along with CGPA:

S.No	Name of the Student	Roll. No	CGPA
1	G.Aishani	24EU04133	8.42
2	B.Tejaswini	24EU04140	8.03
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Title of the project:

AI Driven IV Fluid Monitoring and Smart Alert System

Societal Relevance:

Health Care

Client Details and Location (Societal) with Photograph (paste here):



Abstract:

Intravenous (IV) therapy is a critical component of inpatient clinical care; however, its continuous monitoring remains heavily reliant on manual observation or reactive, threshold-based automated systems. In resource-constrained clinical environments with high patient-to-nurse ratios, this manual reliance exacerbates the risk of severe adverse events, such as air embolisms or undetected venous infiltration. Furthermore, when traditional automated systems are used, they broadcast non-differentiated alarms only after a critical failure has occurred. This reactive approach overwhelms hospital floors with noise and significantly contributes to dangerous "alarm fatigue" among nursing staff. To solve these compounding problems, this project proposes a decentralized, predictive monitoring architecture. By implementing Edge AI directly on an ESP32 microcontroller paired with a Load Cell, the system performs real-time time-series analysis to predict flow anomalies and exact time-to-empty before failures occur. Additionally, a centralized AI Triage Agent ingests this predictive data to dynamically route prioritized, silent alerts directly to the most available staff member. This intelligent allocation system aims to eliminate air embolism risks and reduce broadcast alarm fatigue by up to 40%, safeguarding patients while optimizing nursing workflows.

Keywords

- Remote Patient Monitoring
- Edge Artificial Intelligence
- Multi-Agent Systems
- IoT Medical Devices
- Predictive Analytics

Signature of the EPICS Guide